CSA 1516 - Cloud Computing and Big Data Analytics

CO1 - Short Answers

Sanjay Manesh. M
192511362
B.E CSE

14/10/2026

1. Cloud computing is the delivery of computing resources such as servers, storage, database, networking and software over the internet on demand, usually with pay-as-you-use pricing.

1. On-demand self service: Users can create and access computing resources whenever required without directly containing the service provider
Ex. A company creates a virtual server by AWS.

2. Broad network access: Cloud services are accessible over a network using devices such as laptops, smartphones and tablets.
Ex. Accessing google drive from computer.

3. Rapid elasticity. Resources can be increased or decreased quickly according to workload requirements.

4. Measured service. Resource usage is monitored, customers are charged based on consumption or their chosen subscription.

Traditional computing ~~req~~ requires an organization to purchase, install and maintain its own hardware. Cloud computing provides resources over the internet, reducing the need for upfront hardware investment and allowing flexible scaling.

Conclusion. Cloud computing provides flexible, scalable and accessible computing resources.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution

Cloud computing evolved through several technologies developments. Each development made computing resources easier to share, manage and access remotely.

Evolution of cloud computing.

i, Mainframe computing. Large centralized computers supported multiple users through time sharing

ii, Personal computers. Computing became accessible to individuals and organizations.

iii, Client Server Architecture [1990's]: Processing and data storage ~~was~~ were distributed between clients and servers.

iv, Virtualization [2000's]: Multiple virtual machines could run on a single physical server, improving hardware virtualization.

Internet and Software Application.

i, Internet expansion: High speed networks enabled remote access to computing resources.

ii, Web Applications: Software became accessible through browsers without local installation of softwares in computers.

iii, Distributed and grid computing: Multiple interconnected computers worked together to process large workloads.

iv, Service oriented: Applications were developed using reusable network accessible drivers.

Differentiation between.
IaaS, PaaS, SaaS, XaaS.

1. IaaS - Infrastructure as a Service.

=) IaaS is a service type, which provides fundamental, virtualized IT infrastructure like servers, storage and networking over the internet.

User manages. Operating system, middleware, runtime, applications and data.

Provider manages. Physical hardware, data centers, servers and virtualization layers.

Ex. Amazon Web Services [AWS]
Microsoft Azure, Google compute engines.

2. PaaS - Platform as a Service

=) It delivers a ready to use development and deployment environment so developers can build software without managing the underlying hardware.

User manages. Custom code and applications
Provider manages. OS, database, infrastructure

Ex. Elastic BeanStalk, Google App engine.

3. SaaS - Software as a Service.

=) Delivers complete, fully functional software applications over the web on a subscription basis.

User manages. Account access, user permissions and personal and company data

Differentiation between.
IaaS, PaaS, SaaS, XaaS.

1. IaaS - Infrastructure as a Service.

=) IaaS is a service type, which provides fundamental, virtualized IT infrastructure like servers, storage and networking over the internet.

User manages. Operating system, middleware, runtime, applications and data.

Provider manages. Physical hardware, data centers, servers and virtualization layers.

Ex. Amazon Web Services [AWS]
Microsoft Azure, Google compute engines.

2. PaaS - Platform as a Service

=) It delivers a ready to use development and deployment environment so developers can build software without managing the underlying hardware.

User manages. Custom code and applications
Provider manages. OS, database, infrastructure

Ex. Elastic BeanStalk, Google App engine.

3. SaaS - Software as a Service.

=) Delivers complete, fully functional software applications over the web on a subscription basis.

User manages. Account access, user permissions and personal and company data

Provider manages. Servers, security, maintenance, updates and application

Ex. Google Workspace, Microsoft 365.

4. XaaS. [Everything as a Service]

=> A broad, overarching term for delivering almost any IT capability or specialized business function via the cloud.

User manages. Depending on specific service.

Provider manages. The underlying service.

Ex. Database as a Service [DBaaS], Container as a Service [CaaS].

3. Virtualization (vs) Cloud computing.

	Feature	Virtualization	Cloud computing
i)	Core nature	A technology to abstract hardware	An environment / Service model
ii)	Primary focus	Hardware consolidation and isolation	On demand delivery and scaling
iii)	Automation level	Mostly manual configuration.	Fully automated via APIs and Portals.
iv)	Location	Usually on-premises	Off site / remote network